CACTUS FALL 2026 · WEEK 1 · SESSION 1

Define your problem

By the end of today you will have a project card: a case study, one main question, and a plan for the data.

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THE NEXT EIGHT WEEKS

From a rough idea to working code

Week	Focus
1	Define your problem, and how code thinks (today)
2	Clean it, chart it, check what the AI wrote
3	Verify and analyze: statistics you can test
4	Collect your own data, and the ethics of doing so
5	Analyze text, including Chinese
6	AI as annotator: compare it with human labels
7	Choose a track: networks, images, or a mock feed
8	Reproduce your pipeline and present it

HOW WE WORK

AI writes. You decide.

Describe the task

Read what it wrote

Test it

Record the check

WHY CHECKING MATTERS

Four ways AI code goes wrong

Silent data loss

The code runs, but rows or values quietly disappear or change.

Invented names

Columns, functions, or packages that do not exist.

Does something else

The code answers a different question from the one you asked.

Confident, wrong explanations

It sounds sure of itself. That is not evidence.

HOW A WEEK WORKS

Two sessions, plus a little in between

Learning session

About 2 hours. Lecture, worked examples, and a guided activity.

Coding session

About 2 hours. Breakout rooms and a notebook. No lecture.

In between

Required reading, a short practice page, and your verification log.

All sessions are on Zoom. Bring a laptop. No grades, but each week builds on the last.

STEP 1

Pick a case study

TOO BROAD

“Social media”

A CASE STUDY

How Singapore, Hong Kong, and Taiwan users talk about AI chatbots on Reddit

A case study is a specific topic in a specific context: who it is about, where, and in what period.

STEP 2

One main question, two or three sub-questions

MAIN QUESTION

Can we gauge mental health through social media behavior?

Sub-question 1

What are the main reasons people use social media?

Sub-question 2

Which language behaviors predict low mental health?

Sub-question 3

Do motives change the link between use and mental health?

STEP 3

Pick a structure to start from

Structure	Best for
Predictors, consequences, moderators	A behavior with a clear downstream outcome
Mechanism or mediation	How or why an effect happens
Comparative	Two platforms, places, groups, or periods
Debate	Findings that are visibly mixed
Multi-level	Individual, platform, and policy angles
Problem, cause, intervention	A specific harm and what might fix it

These are starting points, not a checklist. Mix them, or use your own.

A COMPARATIVE EXAMPLE

Starbucks comments across Asia

MAIN QUESTION

How do online comments about Starbucks differ across Southeast and East Asian countries during a boycott period?

How much do comments in each country mention the boycott?

Counting

How does tone differ between countries?

Sentiment

Do the differences hold in every language?

Comparing groups

STEP 4

Is it doable in eight weeks?

Data

Do I have it, or can I get it by Week 4? What is one row?

Size

Big enough to say something, small enough for a laptop?

Measure

For each sub-question, what would I count, label, or compute?

Method

Which week of this series teaches the tool I need?

Ethics

Public data or consent? Could anyone be identified?

Scope

Could I answer it in a few weeks of part-time coding?

Too big: How does social media affect society? **Better:** How do Singapore and Taiwan Reddit users talk about AI-generated art?

STEP 5

Use AI to sharpen, not to decide

```
Here is my main question and sub-questions: (paste). Which structure fits best, and
why? What is missing? What could I not answer with the data I described?
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Then check it. Does the suggestion fit your data? Is the match real, or just plausible-sounding? Write down what you accepted and what you rejected.

YOUR TASK NOW · 35 MINUTES

Draft your project card

YOU FINISH WHEN YOU HAVE

- A case study (topic, place, period)
- One main question and two or three sub-questions
- What one row of your data is, and where it comes from
- One thing to measure for each sub-question

HOW IT RUNS

- Pairs in breakout rooms
- Use AI to sharpen, and log what you accept and reject
- No data yet? Use the Starbucks or Lululemon file
- Facilitators visit and sign off

A FIRST LOOK AT CODE

What is an API?

YOUR ORDER

Parameters

What to search for, which country, which language.

THE KITCHEN

The service

SerpAPI, which asks Google for you.

THE DISH

The response

Labeled, nested data, usually called JSON.

You do not walk into the kitchen. You send an order in a fixed format, and the dish comes back.

NEXT SESSION

How code thinks

In the session

A drag-and-drop puzzle page, then the Colab notebook. No lecture, breakout rooms.

Before then

Read Chapter 3 of the textbook (sections 3.1 to 3.3). Finish your project card.

Optional

Create a free SerpAPI account. Saved searches work for everyone, so this is not required.

Textbook: Computational Analysis of Communication, free at cssbook.net